

Bowen Song

Embodied / Biomechanics / Interpretable Time-Series

Research engineer building interpretable wearable ML that survives noisy sensing and edge limits.

Embodied sensing is noisy, personalized, and resource-constrained, so I build interpretable multimodal models that can leave the lab and run under edge limits.

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EXPERIENCE

Graduate Researcher, Personalized AI for Health

2020-Present

University of Southern California | Los Angeles, CA

Wearable sensing is noisy and deployment is constrained, so I built multimodal pipelines and personalized models that stay interpretable on device and useful across users.

- Wearable force estimation needed a smaller baseline, so I led SER and showed low-rank models could match or beat heavier LSTMs while staying interpretable and credible for edge deployment.
- Research pipelines were fragmented across sensing modalities, so I built end-to-end data, training, and evaluation workflows that made comparison under noise and subject shift reliable enough to guide model decisions.

Researcher, Cloud Resource Management

2020-2022

University of Southern California | Los Angeles, CA

Resource sharing across clouds is hard to trust, so I modeled QoS and incentive constraints to show when federation actually improves system utility and adoption.

- Hybrid-cloud federation can shift load without helping users, so I built a queueing framework that identified when QoS-constrained sharing improves utility enough to be worth deploying.

PhD Research Intern, Core Ranking

2024

eBay | San Jose, CA

Ranking gains had to survive real traffic and strong baselines, so I built production-constrained model and evaluation workflows that made improvements worth trusting.

- Model novelty can hide weak measurement, so I carried research-style evaluation discipline into ranking and kept decisions tied to operationally realistic metrics the team could trust.

PUBLICATIONS

Estimating Ground Reaction Forces From Inertial Sensors

IEEE TBME 2024 | B. Song, M. Paolieri, H. E. Stewart, L. Golubchik, J. L. McNitt-Gray, V. Misra, D. Shah

Ground-reaction estimation from IMUs is noisy and resource-constrained, and this paper shows SER can stay interpretable, lightweight, and credible outside the lab.

Performance and Revenue Analysis of Hybrid Cloud Federations with QoS Requirements

IEEE CLOUD 2022 | Bowen Song, Marco Paolieri, Leana Golubchik

Deployable ML inherits systems constraints, and this paper shows the resource and incentive reasoning behind how I think about scaling under limits.

PROJECTS

SER PAPER IMPLEMENTATION

SER

Wearable GRF estimation needed a lighter baseline, so I implemented SER and showed interpretable low-rank modeling could stay accurate enough for deployment-minded biomechanics work.

LARGE-SCALE RECONCILIATION

RCDS_GO

Large remote states are expensive to resync, so I implemented content-aware reconciliation that sends differences instead of whole payloads and makes huge-file sync more practical.

EDUCATION

PhD in Computer Science

University of Southern California | Los Angeles, CA | 2020-2026 (expected)

MS in Electrical and Computer Engineering

Boston University | Boston, MA | 2019

SKILLS

Modeling: Multimodal and time-series modeling for noisy wearable sensing and interpretable low-rank inference; Federated personalization, subject adaptation, and edge-aware model design for heterogeneous users

Systems: Cloud federation, QoS reasoning, benchmarking, and reproducible infrastructure for scalable shared systems